

Ayush Kumar Gupta

BTech in Computer Science | MS in Computational Linguistics
International Institute of Information Technology Hyderabad (IIIT-H)
akgiith@gmail.com | +91-9993345988 | LinkedIn

Education

International Institute of Information Technology Hyderabad (IIIT-H) July 2023 – Aug 2028
BTech in Computer Science | MS in Computational Linguistics

CGPA: 7.9

Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing, Information Retrieval, Data Structures & Algorithms, Operating Systems, Probability, Linear Algebra.

Research Experience

Undergraduate Researcher – MT-NLP Lab, LTRC, IIIT Hyderabad Apr 2025 – Present
Supervisor: Prof. Manish Shrivastava

Focusing on **mechanistic and developmental interpretability** in neural models. Investigating internal representations, attention dynamics, and systematic failure modes to understand how models learn linguistic phenomena and acquire knowledge from indirect supervision.

Teaching Experience

Teaching Assistant – Introduction to Natural Language Processing (INLP) Jan 2026 – Present
IIIT Hyderabad

Designing assignments, conducting tutorials, and mentoring students on NLP fundamentals including Language Modeling, Syntax Parsing, and architectures for Sequence Labeling tasks.

Industry Experience

Software Development Intern – Kaizen Platforms Jan 2025 – Apr 2025

Contributed to backend development for *Finease*, a production-grade application. Focused on scalable logic, feature integration, and debugging while following structured software engineering workflows.

Projects

Impact of Tokenization on Machine Translation (*Research Project*) Investigating the correlation between vocabulary size (BPE, Unigram) and NMT metrics. Analyzing how segmentation choices impact translation consistency across similar language pairs.

Machine Unlearning in VLMs (*Course Project*) Evaluated targeted forgetting algorithms in VLMs to erase specific knowledge subsets using SAEs.

Subliminal Learning and Token Entanglement (*Research Project*) Studied representational leakage and token entanglement in LMs. Analyzed implicit concept learning under controlled prompting to provide insights into knowledge acquisition from indirect supervision.

Information Retrieval System (*IRE Assignment & Project*) Built a full IR pipeline (crawler, indexer, ranker) from scratch. Compared TAAT and DAAT scoring strategies against Elasticsearch benchmarks for retrieval efficiency and ranking quality.

Sequence Labeling for Indian Languages (NER & ABSA) Developed CRF-based systems for Hindi NER and Aspect-Based Sentiment Analysis. Optimized linguistic feature engineering and context-window selections to enhance structured prediction accuracy.

Technical Skills

Languages: Python, C, Bash

Tools: Git, Linux, Elasticsearch, Pandas, LaTeX

ML/DL: PyTorch, NumPy, scikit-learn, Transformers

NLP Areas: Interpretability, Machine Translation, Tokenization, Information Retrieval, Embeddings

Prompting: Prompt Engineering, Chain-of-Thought, Model Evaluation, Agentic Systems

Research Interests

Mechanistic Interpretability, Developmental Interpretability, Singular Learning Theory, Deep Learning, Explainable AI, Multimodal NLP